



SAFETY INDUCTION

CNPC Chuangqing Drilling Engineering Company Limited



Welcome

Welcome to CCDC-73

This safety induction is designed to ensure your visit is safe and compliant with our health and safety standards

Our priority is to protect you, our employees, and the environment



OUR MAIN GOALS

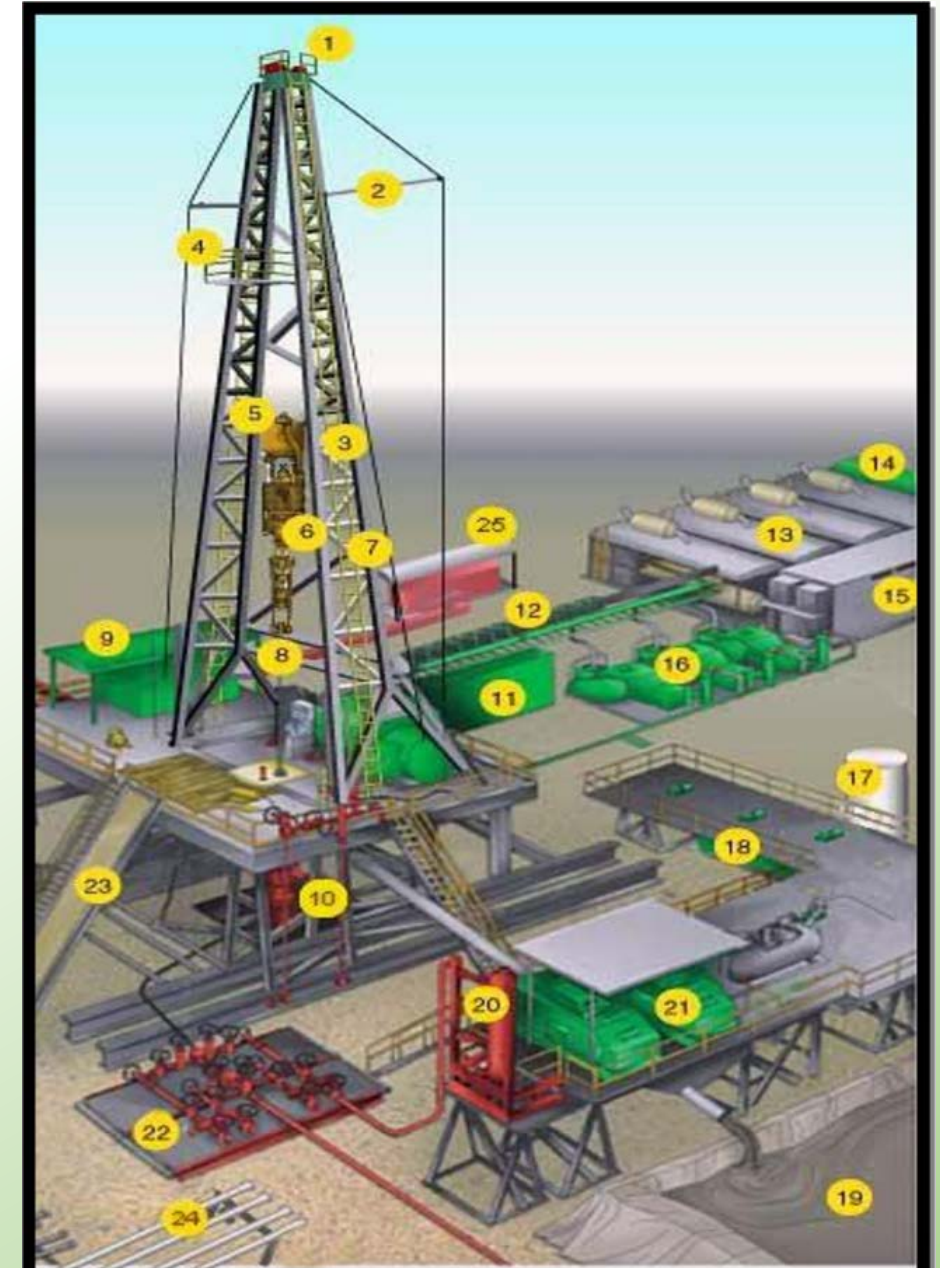
- Where safety is our top priority. Our QHSE principles guide everything we do here
- We're committed to maintaining the highest standards in Safety, Health, Environment, and Quality.
- Our target is zero incidents. This isn't just a goal—it's our expectation for everyone on this rig.



Common Rig Site Equipments

1. Crown Block
2. Catline Boom and Hoist Line
3. Drilling Line
4. Monkey board
5. Traveling Block
6. Top Drive
7. Mast
8. Drill Pipe
9. Doghouse
10. Blowout Preventer
11. Water Tank
12. Electric Cable Tray
13. Engine Generator Sets
14. Fuel Tank
15. Electrical Control House
16. Mud Pumps
17. Bulk Mud Component Tanks

18. Mud Tanks (Pits)
19. Reserve Pit
20. Mud-Gas Separator
21. Shale Shakers
22. Choke Manifold
23. Pipe Ramp
24. Pipe Racks
25. Accumulator





HAZARD ZONE CLASSIFICATION

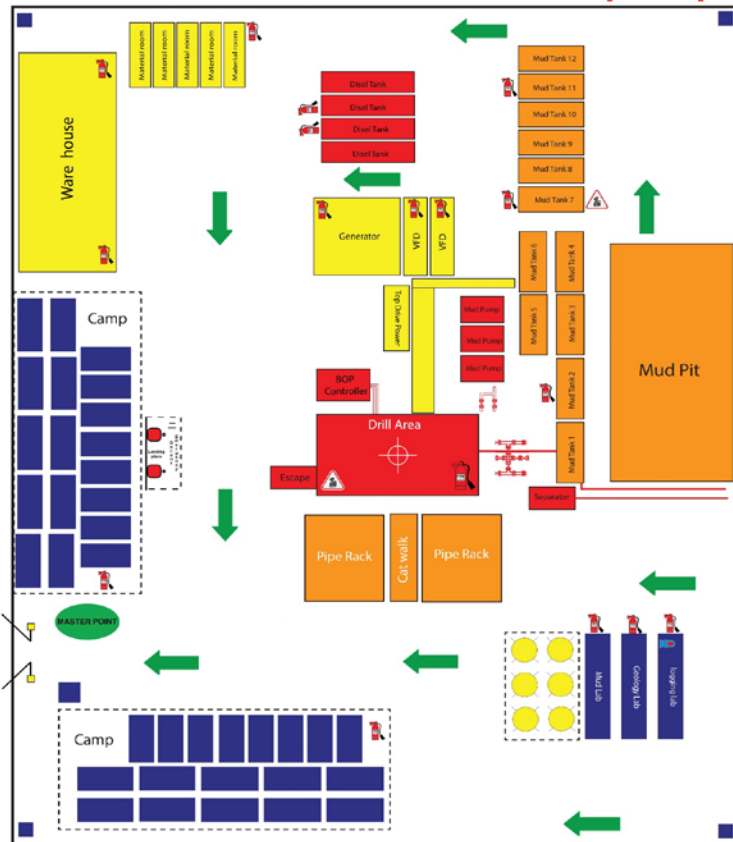


中国石油川庆钻探工程公司

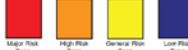
CNPC CHUANQING DRILLING ENGINEERING COMPANY

井场风险分级及逃生示意图

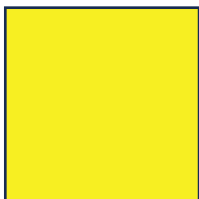
Risk Classification & Evacuation Escape Map



LEGEND



General Risk
Zone



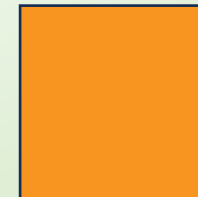
Low Risk
Zone



Major Risk
Zone



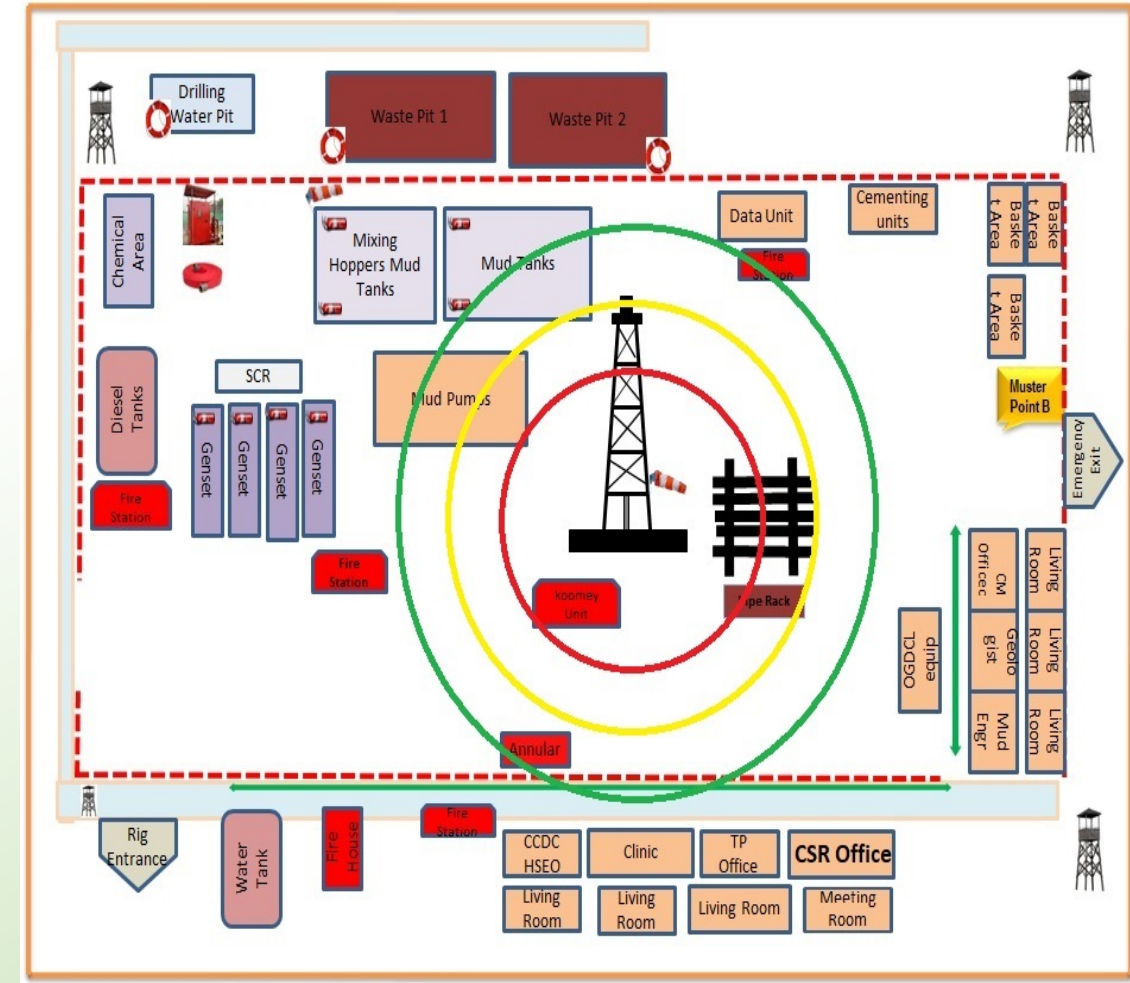
High Risk
Zone





Rig Site Hazard Zone Classification

- Red Zone (0–15 meters): Defined as the Critical Zone. This is where the highest concentration of mechanical and pressure-related hazards exists.
- Yellow Zone (15–25 meters): Defined as the Caution Zone. This area requires increased situational awareness and specific PPE.
- Green Zone (Beyond 25 meters): Defined as the Safe Working Zone. This is the distance required to protect non-essential personnel from the rig's primary operational hazards.





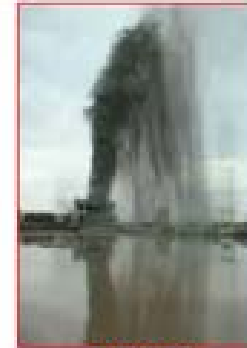
Specific Hazard & Risk



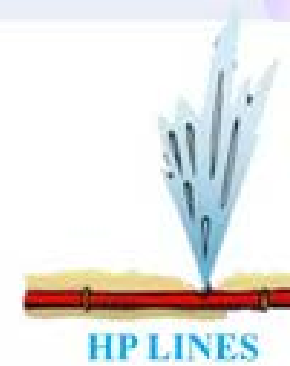
FIRE



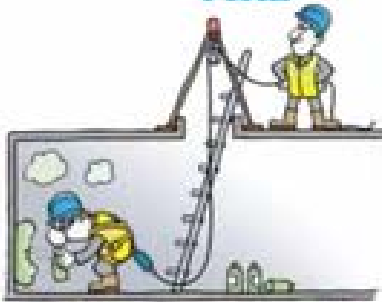
TOXIC GAS



KICK



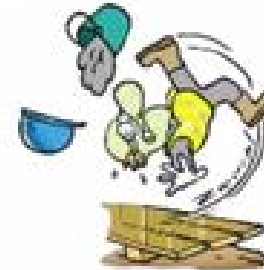
HP LINES



CONFINED SPACE



LIFTING ACTIVITIES



SLIP / TRIP



MOVING EQUIPMENTS



BIOLOGICAL



MANUAL HANDLING



WORK AT HEIGHT



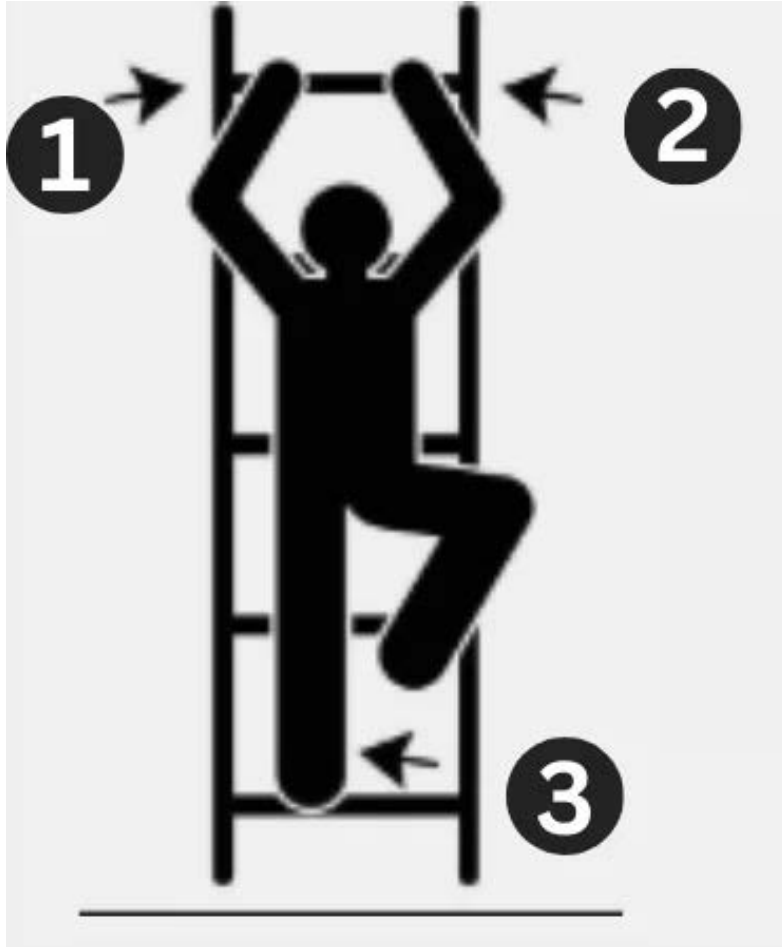
Good Safety Habits on Site



⚡ Discharge Static Electricity – Always touch the static discharge plate before climbing stairs or entering equipment areas.



Good Safety Habits on Site



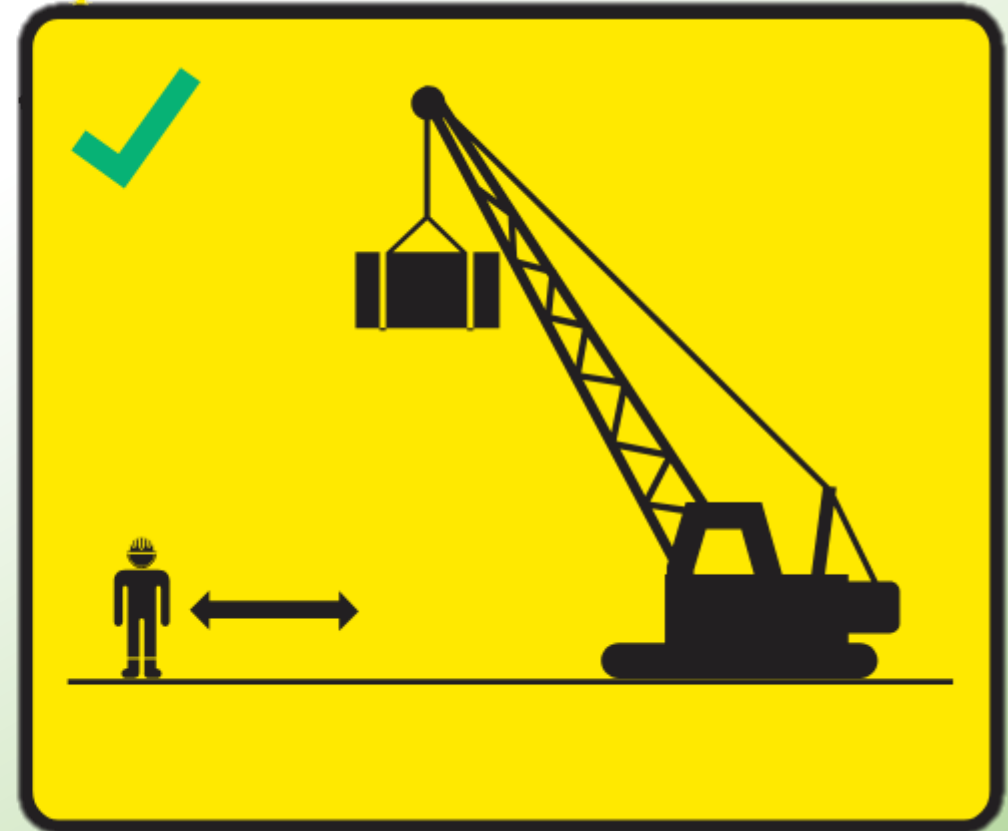
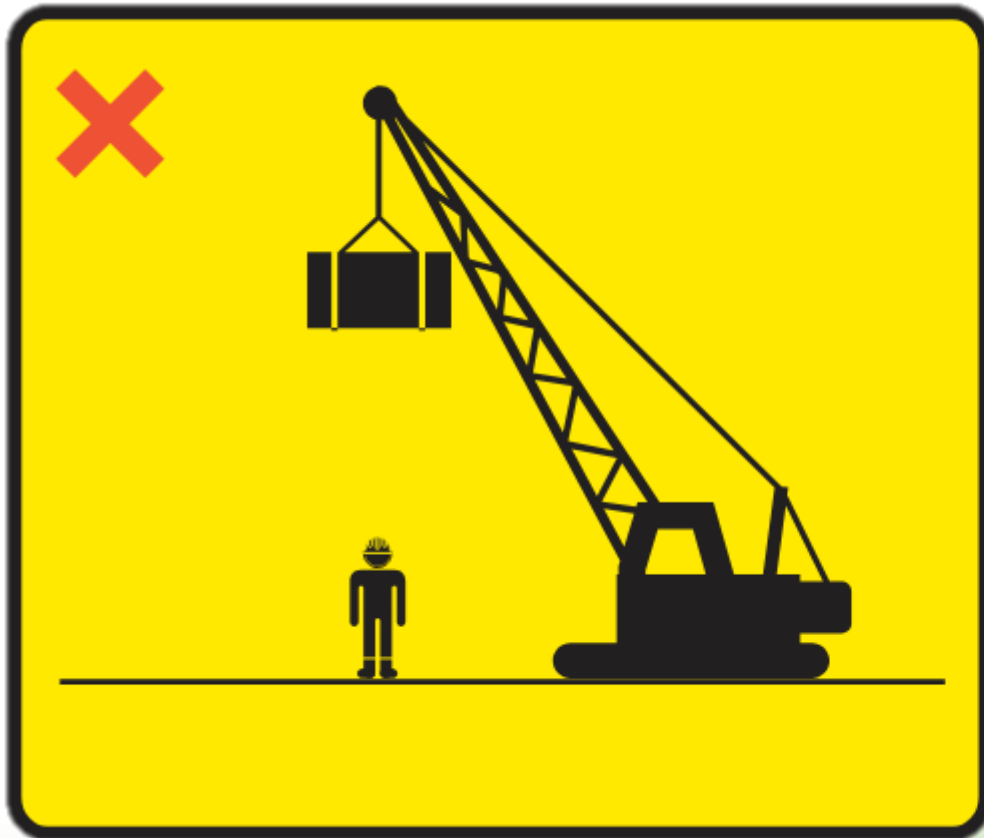
Use 3-Point Contact— while climbing stairs or ladders.

- 2 hands , 1 foot
- 1 hand 2 feet



Good Safety Habits on Site

Do not stand under suspended load at all times





Good Safety Habits on Site

Beware of

SLIP



Trip



Fall





Good Safety Habits on Site



Restricted Area:– Do not enter areas marked with barrier tape.

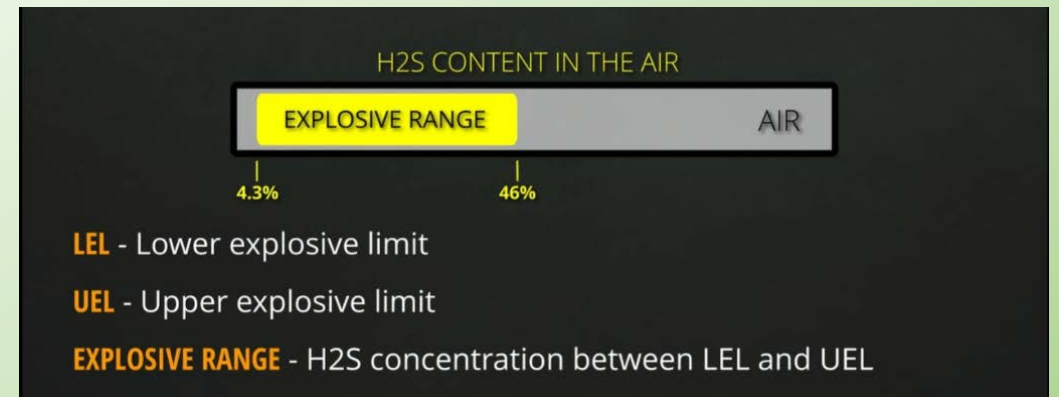


H₂S Awareness

H₂S (Hydrogen Sulfide) is a highly toxic, flammable gas commonly found in oil & gas operations.

Characteristics:

- Colorless (can't see it)
- Smells like a rotten egg
- Heavier than air (run away upwards if you smell it)
- Common in low-lying areas with little air movement (sewer gas)
- Explosive, flammable





H2S Awareness

EXPOSURE LIMITS

0 - 10 ppm is "not dangerous" - for limited time period
(but it would be better to wear a SCBA).

10 - 15 ppm is serious and you should get out of there ASAP.
(assuming you are not wearing SCBA)

15 - 20 ppm is very dangerous and you MUST have a SCBA to be there

Everything more than that can seriously harm you,
depending on **how long** you have been exposed
and to **which concentration** level of H2S in the air.

IDLH - 100 ppm



CEL - 20 ppm —

STEL - 15 ppm —

TWA - 10 ppm —



Hazard Reporting

THE STOP SAFETY OBSERVATION CYCLE

Decide  Report

MAKE IF ANY UNSAFE  OBSERVE  MAKE IF ALL SAFE

OBSERVATION CHECKLIST

REACTIONS OF PEOPLE ☐

- ☐ Adjusting Personal Protective Equipment
- ☐ Changing Position
- ☐ Rearranging Job
- ☐ Stopping Job
- ☐ Attaching Grounds

PERSONAL PROTECTIVE EQUIPMENT ☐

- ☐ Head
- ☐ Eyes and Face
- ☐ Ears
- ☐ Respiratory System
- ☐ Arms and Hands
- ☐ Trunk

POSITIONS OF PEOPLE (Injury Cause) ☐

- ☐ Striking Against Objects
- ☐ Struck By Objects
- ☐ Caught In, On, or Between
- ☐ Objects
- ☐ Falling
- ☐ Contact Temperature Extremes
- ☐ Contacting Electrical Current
- ☐ Inhaling
- ☐ Absorbing
- ☐ Swallowing
- ☐ Overexertion
- ☐ Repetitive Motions

TOOLS AND EQUIPMENT ☐

- ☐ Postures
- ☐ Wrong for the Job
- ☐ Used Incorrectly
- ☐ Unsafe Condition

PROCEDURES AND ORDERLINESS ☐

- ☐ Procedures Not Adequate
- ☐ Procedures Not Known/Understood
- ☐ Procedures Not Followed
- ☐ Orderliness Standards Inadequate
- ☐ Orderliness Standards Not Known/Understood
- ☐ Orderliness Standards Not Followed

STOP **OBSERVATION REPORT**

• SAFE ACT OBSERVED
• ACTIONS TAKEN TO ENCOURAGE CONTINUED SAFE PERFORMANCE

• UNSAFE ACTS OBSERVED
• IMMEDIATE CORRECTIVE ACTION
• ACTION TO PREVENT REOCCURRENCE

Observer's Signature _____
Area/Dept. _____ Date _____

© DuPont Company K15-04-124

- Empowerment to STOP the job
- STAR Cards shall be used to report all safe/unsafe conditions and acts.
- Report any Hazards or Incidents to the HSE and Rig Manager immediately.



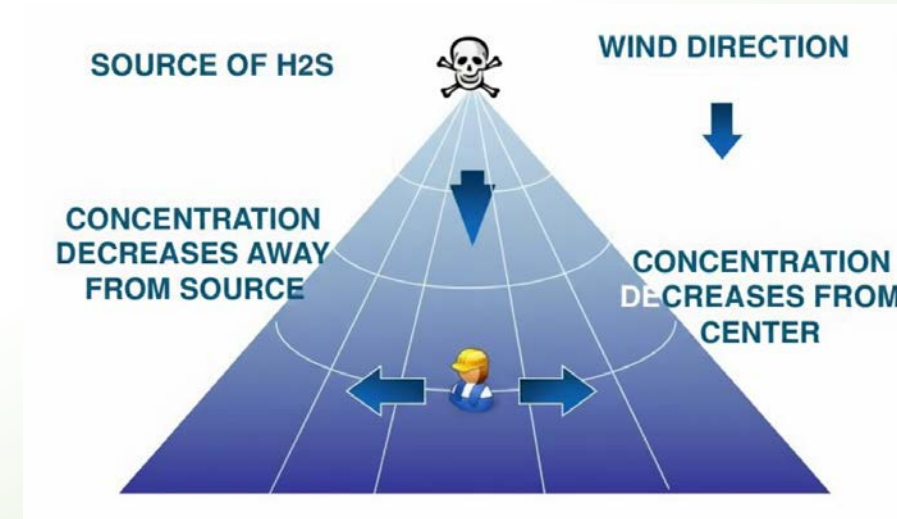
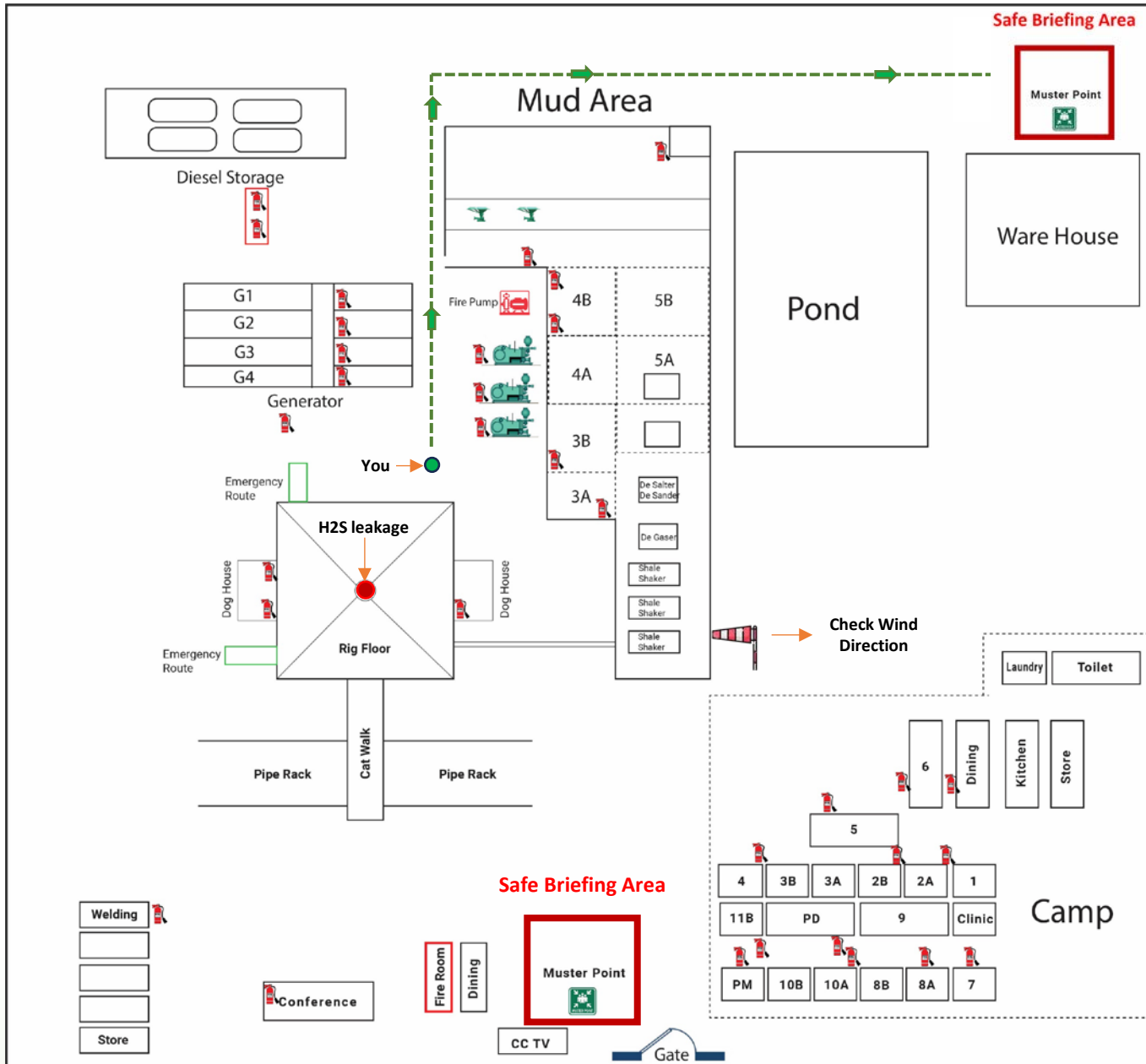
Permit to Work

- Certain high-risk jobs cannot be done without a permit, such as hot work, confined space entry, electrical work, lifting operations, and working at height.
- Always check if a permit is in place before starting any task. Never begin work until your supervisor explains the permit and your role..
- Do not enter or approach areas where permit-controlled jobs are ongoing unless escorted by an authorized person.

Distribution: 1st copy -Permit Receivers (Must be displayed at work site), 2nd copy - Permit Issuer, 3rd copy - Permit Receivers



Emergency Escape Route



GO CROSS-WIND OR UP WIND



Rig Emergency Alarm Signals 应急警报信号

● Kick / Blowout 井控：连续长鸣 ≥15 秒

Continuous siren for more than 15 seconds.



Continuous siren (more than 15 seconds)

● H2S / Toxic gas 硫化氢：两短一长 2s-2s-15s

Two short blasts followed by one long blast (2s – 2s – 15s)



2s blast 2s blast 15s blast

● Armed Attack 武装袭击：1秒短鸣连续15次以上

Repeated 1-second short blasts (15 or more repetitions)



1s blast 1s blast 1s blast 1s blast 1s blast 1s blast 1s blast 1s blast 1s blast 1s blast 1s blast

● Fire 火灾：长鸣10秒 - 停2秒 - 长鸣10秒

Two 10-second long blasts separated by a 2-second pause.



10s blast 2s pause 10s blast

● Medical emergency 医疗急救：通过对讲机呼叫

Report by radio only (No siren signal).



Emergency Contact





Thank You

Have a Safe Stay

